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| Semester: III                           | Year : AY 2026-27 |
| Subject: Programming with Java          |                   |
| ASSIGNMENT No : 10                      |                   |
| TITLE : Abstract Classes and Methods    |                   |

### Problem Statement

Write a Java program to demonstrate the use of abstract classes and abstract methods

### Learning Objectives

To implement abstraction using abstract classes and abstract methods.

### Programme

```
abstract class Shape
{
    abstract void draw();
    void message()
    {
        System.out.println("this a shape. ");
    }
}

class Circle extends Shape
{
    void draw()
    {
        System.out.println("drawing a cercle ");
    }
}

public class AbstractExample
{
    public static void main(String[] args)
    {
        Shape s= new Circle();
        s.draw();
        s.message();
    }
}
```

## Output

```
PS C:\CS-JAVA> javac AbstractExample.java
PS C:\CS-JAVA> java AbstractExample
drawing a cercle
this a shape.
PS C:\CS-JAVA> 
```

## Conclusion

In conclusion, an **abstract class** is an incomplete class that is used as a base class for other classes. An **abstract method** is a method without a body, and the child class must provide its implementation. Abstract classes can also contain normal methods with complete implementations. We cannot create an object directly from an abstract class. They are useful when we want to provide a common structure to different child classes while allowing each child to define its own behavior.